COURSE TITLE : **PULP TECHNOLOGY**
COURSE CODE : **3122**
COURSE CATEGORY : **B**
PERIODS PER WEEK : **4**
SEMESTER : **3**
PERIODS PER SEMESTER : **60**
CREDITS : **4**

TIME SCHEDULE

Module	Topics	Periods
I	Preparation of Raw materials Chemical Engineering Test	7 7 1
II	Mechanical Pulping Semi chemical pulping Pulp from Secondary Fibres Test	5 5 4 1
III	Washing of Pulp Pulp Screening and Cleaning and Screening Test	7 7 1
IV	Bleaching of Pulp Pulping with Agricultural residues Test	7 7 1
	Total	**60**

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand The Preparation of raw materials For pulping
	2	Comprehend Chemical Pulping
2	1	Understand Mechanical Pulping Process
	2	Understand Semichemical Pulping Process
	3	Understand Pulp From Secondary Fibers
3	1	Comprehend Washing of Pulp
	2	Understand Screening and Cleaning of Pulp
4	1	Understand Bleaching of Pulp
	2	Comprehend Pulp from agricultural residue

Specific outcome

MODULE I

1.1.0 Understand The Preparation of Rawmaterials For Pulping

1.1.1 Define Pulp, paper and Paper
1.1.2 Describe the historical development of pulp and paper Industry and its relevance
1.1.3 Explain the procurement of Various raw materials for Pulp Industry
1.1.4 Differentiate Chemical Pulping, Mechanical pulping and semi chemical Pulping
1.1.5 state Debarking of wood
1.1.6 Explain debarking machines
1.1.7 Distinguish between drum chippers and disc chippers
1.1.8 Explain chippers with Diagramme
1.1.9 Describe chip conveyors and chip silo

1.2.0 Comprehend Chemical Pulping

1.2.1 State Chemical pulping
1.2.2 Distinguish between Chemical Pulping, Mechanical Pulping and semi chemical pulping
1.2.3 Explain Sulphate Pulping Process
1.2.4 Describe the Preparation of White Liquor
1.2.5 Explain batch pulping
1.2.6 State chip Liquor ration
1.2.7 Estimate chip liquor ratio
1.2.8 Identify continuous pulping
1.2.9 Describe Pandya Continuous Digester, Kamyr Digester and Medura repulper with Figure
1.2.10 Compare batch digester with continuous Digester

MODULE II

2.1.0 Understand Mechanical Pulping Process

2.1.1 Define mechanical pulping
2.1.2 Compare pulping process
2.1.3 State the principles mechanical pulping
2.1.4 List Mechanical pulping
2.1.5 Define ground wood pulping
2.1.6 Explain grinders with figure
2.1.7 Describe G.W pulping
2.1.8 State Refiners
2.1.9 Explain refiners
2.1.10 Describe variables in refining
2.1.11 Explain thermo mechanical and & CTMP

2.2.0 Understand Semi-chemical Pulping Process
2.2.1 State principles of semi chemical pulping
2.2.2 List various semi chemical process

2.2.3 Explain chemi- mech pulping
2.2.4 Justify latest development in pulping process
2.2.5 State Super batch cooking
2.2.6 State RDH(Rapid Displacement of Heat) Cooking
2.2.7 State Oxygen delignification

2.3.0 Understand Pulp From Secondary Fibers
2.3.1 Define secondary fibers
2.3.2 Give Examples of various secondary fibers
2.3.3 Explain the production of secondary fibers
2.3.4 State the importance of recycling
2.3.5 Explain deinking process
2.3.6 Describe floatation cell
2.3.7 Describe properties and uses of deinked paper
2.3.8 Describe effluent treatment and waste disposal in Deinking plant
2.3.9 Explain Depithiong methods
2.3.10 Explain pulping Bagasse
2.3.11 Explain the properties and uses of bagasse pulp
2.3.12 State rayon grade pulp
2.3.13 Describe the method of production of rayon grade pulp

MODULE III

3.1.0 Comprehend Washing of Pulp

3.1.1 Define pulp washing
3.1.2 State the objective of washing
3.1.3 Describe rotary drum vacuum washer with flow diagram
3.1.4 Define pressure washer
3.1.5 State high consistency washing
3.1.6 List presses use in pulp washing
3.1.7 Explain screw press
3.1.8 Explain belt press
3.1.9 Explain disc presses

3.2.0 Understand Screening and Cleaning of Pulp

3.2.1 State the need of screening
3.2.2 List the screens
3.2.3 Explain flat screens with sketches
3.2.4 Explain rotary screens
3.2.5 Explain pressure screen, centrifugal screens
3.2.6 State high consistency screens
3.2.7 List century cleaners
3.2.8 Explain vertical cleaner
3.2.9 Explain horizontal cleaner
3.2.10 Explain inclined cleaner

MODULE IV

4.1.0 Understand Bleaching of Pulp

4.1.1 Define bleaching
4.1.2 State the need of bleaching
4.1.3 List various stages of bleaching
4.1.4 List Give examples of the various bleaching chemical
4.1.5 Explain important parameter on bleaching – temperature – time – pH – Consistency
4.1.6 List bleach plant equipment
4.1.7 Describe chlorine mixture, steam mixture, chemical mixture, hip mixer, Cautic mixture , retention tower, high density tower
4.1.8 Explain chlorine bleaching, chlorine dioxide bleaching, hydrochlorate bleaching
4.1.9 Explain multi stage bleaching
4.1.10 Explain elemental chlorine free bleaching and total chlorine free bleaching
4.1.11 Describe the preparation of bleaching chemicals

4.2.0 Comprehend Pulp From Agricultural Residue

4.2.1 State the importance of agricultural residue in Pulp industry
4.2.2 List the agricultural residue
4.2.3 Explain the Pulping of rags, straws, cereal straw, esparto, hemp, jute, flax, cotton linters, and bamboo

CONTENT DETAILS

MODULE I

Preparation of Raw Materials for Pulping

Raw materials – sources - classification of raw materials – selection of raw materials - transportation – debarkers – chipping – drum chipper – disc chipper – calculation of chip size – screening of chips – screens - shaking screens – vibrating screens – grizzles – bucket elevators - sieve analysis – re chipper – conveying of chips – belt conveyor – chain conveyors, roller conveyor, pneumatic conveyors scrap conveyor –screw conveyor - advantageous & disadvantageous - Chip storage - chip silo

PULPING PROCESS

Classification of pulping process – Chemical pulping – Soda pulping – sulphite process – sulphate pulping - bisulphite process- Principles of chemical pulping – soda process – sulphate process - sulphate pulping –bisulphate process-Bath ratio- Active alkali – effective alkali – sulphidity- consistency- TTA-kappa number- permanganate number- cooking process- Digester- Classification- batch process- Direct & indirect heating- Continuous cooking- Kamyr- Pandya digester- rotary Digester

MODULE II

Mechanical pulping and semi-chemical pulping.Principles ofmechanical pulping- comparison of pulping processes. Types of mech.pulping- Ground wood pulping-. Grinders –types of grinders- Refiners-variables in refining. Thermomechanical pulping, Chemi thermomechanical pulping. Principles of semi-chemical pulping. Different semi-chemical pulping processes. Chemi-mech pulping. Super batch cooking. RDH cooking-Oxygen delinification.

Pulp from secondary fibers

.Definition-classification of secondary fibers- Production of secondary fibers. Importance of recycling . Deinked pulp –advantages –economics of deinking. Classification of ink -.Deinking process. Variables in deinking operation. Flotation cell. Dipergent of deinked stock –properties and uses of deinked paper. Effluent treatment and waste disposal in deinking plant. Depithing – classification of depithing methods. Pulping of bagasse- properties and uses of bagasse pulp

MODULE III

Washing of pulp – objectives and principles of washing. Rotary drum vacuum washer,pressur washer .— High consistency washing –latest development in washing. Presses used in washing –Screw press –disc press –belt press . Screening and cleaning of pulp. Need of screening and cleaning. Types of screens – rotary screen –pressure screen – centrifugal screen – high consistency screen. Centry cleaners – vertical and horizontal cleaners – inclined cleaner – high consistency cleaner

MODULE IV

Bleaching of pulp. Bleaching – need of bleaching –theory of bleaching –principles of bleaching . Various stages in bleaching . Types of chemicals –parameters- temp. ,time ,PH , consistency . Equipments in bleach plant. Types of mixers. Types of bleaching – chlorine , chlorine dioxide , hypochlorate bleaching. Multi stage bleaching ,oxygen bleaching ,ozone bleaching- Total chlorine free bleaching, Elemental chlorine free bleaching. Prepartation of bleaching chemicals . Productive bleaching . Cooking of agricultural residue –importance of agricultural residue . Preparation and pulping of rags ,rice ,straw ,ceral straw , esparto ,hemp ,jute ,flax ,bamboo , cotton linter .

REFERENCE

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8. Libby C.E., Pulp & paper science Vol. I, II, - McGraw Hill, Newyork
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